

Alexandre Autran

Student in mathematics and machine learning

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Profile

I am a mathematics and machine learning student at the École Normale Supérieure (ENS) in Rennes and Paris-Saclay, interested in the mathematical foundations of machine learning, with a focus on probabilistic methods, high-dimensional statistics, stochastic processes and optimal transport.

Education

2024–2027 Normalien – ENS Diploma in Mathematics

Admitted to the ENS through a competitive entrance examination.
Four-year diploma completed in three years (accelerated track).

2026–2027 Research Year in Machine Learning, École Normale Supérieure Paris-Saclay, France

Final year of the ENS Diploma (research track).

2025–2027 MSc in Mathematics, École Normale Supérieure de Rennes, France

M2 Fundamental Mathematics. Highest honors (17/20 in the first semester).
M1 Fundamental Mathematics. Highest honors (17.6/20). Rank: 5/100+.

2024–2025 BSc in Mathematics, Computer Science minor, École Normale Supérieure de Rennes, France

L3 Fundamental Mathematics. High honors (15.4/20).

2022–2024 BSc in Mathematics, Computer Science minor, University of Rennes, France

L2 Mathematics. Highest honors (18/20). Rank: 3/100+.
L1 Mathematics. Highest honors (18.5/20). Rank: 1/100+.

Experiences [\[Full list here\]](#)

Sep 2026– May 2027 Machine Learning Researcher

High-Dimensional Density Analysis via Local Unimodality.
Supervised by Argyris Kalogeratos (Senior Researcher) at Centre Borelli, ENS Paris-Saclay, France.

May–Aug 2026 M1 Research Internship

Optical Modeling of VO₂-Based Thermochromic Oxide Nanocomposites.
Supervised by Nguyen Ngoc Duy (Prof.) and Amaury Baret (PhD Student) at the Univ. of Liège, Belgium.

Apr–Jun 2026 Research Internship

Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems.
Supervised by Martin Gander (Prof.) and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.

Nov–Dec 2025 M2 Research Seminar

Point Processes and Stein's Method.
Supervised by Jean-Christophe Breton (Prof.) at the University of Rennes, France.

Jun–Aug 2025 L3 Research Internship

Lattice-Based Post-Quantum Cryptography.
Supervised by Pierre-Alain Fouque (Prof.) and Aurore Guillevic (Research Scientist) at Inria, France.

May–Jun 2025 Research Internship

Algebraic Counting of the Real Roots of Polynomials.
Supervised by Marie-Françoise Roy (Prof. Emeritus) at the University of Rennes, France.

Research [\[Full list here\]](#)

- [1] **A. Autran**, Singular Tracking of Fractional Brownian Motion under Proportional Costs, preprint, 2026.
- [2] **A. Autran**, Improved Wasserstein Estimates for Jump-Diffusion Perturbations under Semigroup Smoothing, preprint, 2026.
- [3] **A. Autran**, Optimization with an Exact Discrete Adjoint for a Diffusive Lotka–Volterra Harvesting Problem, preprint, 2026.
- [4] **A. Autran**, D. Munteanu, and J.-L. Autran, Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles, *Applied Sciences*, vol. 16, no. 11, Art. no. 5551, 2026.

Other Activities [Full list here]

- 2024–2025 **ENS Oral Examinations**
Supervision of second-year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures (Ulm, Paris-Saclay, Rennes, and Lyon).
- 2022–2025 **Teaching in Mathematics**
Tutoring for final-year high school students at Lycée Chateaubriand in Rennes, as well as first- and second-year BSc students at the University of Rennes.
- 2023–2024 **BSc Student Representative**
Representative on the council of the Mathematics Department of the University of Rennes.
- Before 2022 **High School Competitions**
Participation in the Concours Général in Mathematics, Physics, and Philosophy, as well as the National High School Mathematics Olympiad, at Lycée Thiers in Marseille.

Computer Skills

Programming: Python (NumPy, pandas, scikit-learn, PyTorch), C/C++, Java, MATLAB, R.

Languages

French: native language. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training.
Interests: fantasy and science fiction literature; chess.

Academic Reference

François Bolley Email: francois.bolley@ens-rennes.fr.
Full Professor and Director of the Mathematics Department at ENS de Rennes.